Why should it be called so? Because one can draw a circle exactly passing through the three vertices of the triangle, with centre at the point of concurrence of the perpendicular bisectors of sides. Thus, the circumcentre is equidistant from the vertices of the triangle.

Activity

Check if the following are true by paper-folding:

1. The circumcentre of an acute angled triangle lies in the interior of the triangle.
2. The circumcentre of an obtuse angled triangle lies in the exterior of the triangle.
3. The circumcentre of a right triangle lies at the midpoint of its hypotenuse.

Example 5.19

In ΔABC, S is the circumcentre, BC = 72 cm and DS = 15 cm. Find the radius of its circumcircle.

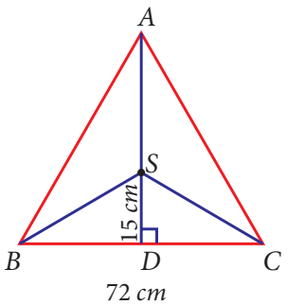


Fig. 5.36

Solution:

As S is the circumcentre of ΔABC, it is equidistant from A,B and C. So AS=BS=CS=radius of its circumcircle. As AD is the perpendicular bisector of BC, $BD = \frac{1}{2} \times BC = \frac{1}{2} \times 72 = 36$ cm

In right angled triangle BDS, by Pythagoras theorem,

$BS^2 = BD^2 + SD^2 = 36^2 + 15^2 = 1521 = 39^2 \Rightarrow BS = 39$ cm.

∴ The radius of the circumcircle of ΔABC is 39 cm.

5.9 Angle Bisectors of a Triangle

We have learnt about angle bisectors in the previous class. An **angle bisector** is a line or ray that divides an angle into two congruent angles. In the figure, ∠ABC is bisected by the line BD such that ∠ABD = ∠CBD.

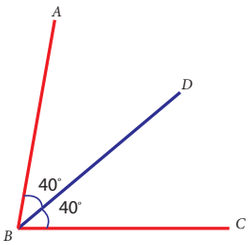
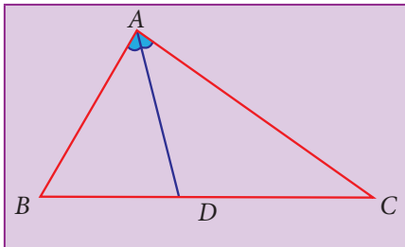
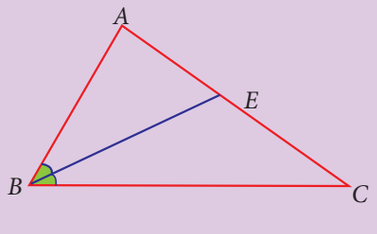
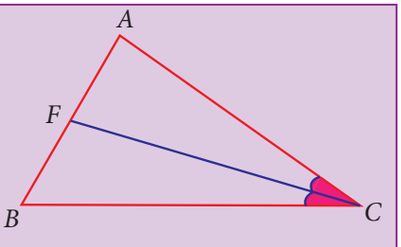


Fig. 5.37

Consider a triangle ABC. How many angles does a triangle have ? 3 angles. For each angle you can have an angle bisector as follows:

		
AD bisects ∠A into two congruent angles. Hence it is an angle bisector of ∠A.	BE bisects ∠B into two congruent angles. Hence it is an angle bisector of ∠B.	CF bisects ∠C into two congruent angles. Hence it is an angle bisector of ∠C.

8th_Maths_English Medium_Chapter 5.indd 175 23/11/2022 15:49:57